Activity-1

20/7/26

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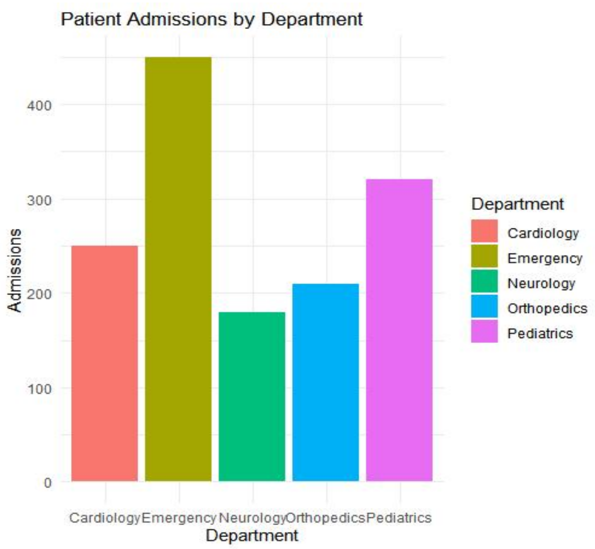
1. A hospital has collected data on patient admissions across multiple departments(Cardiology, Neurology, Pediatrics, Orthopedics, Emergency) for the past year.Different visualizations (bar chart, stacked bar chart, treemap, box plot, histogram) are available. Evaluate which visualization most effectively communicates departmentworkload, admission variability, and patient distribution. Justify your selection withevidence.

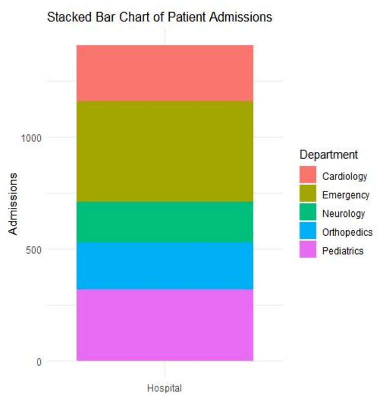
Code:

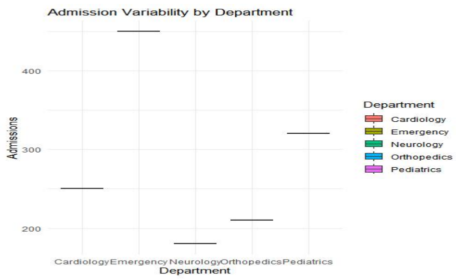
```
library(ggplot2)
library(dplyr)
library(treemapify)
hospital_data <- data.frame(
  Department = c("Cardiology","Neurology","Pediatrics",
                 "Orthopedics","Emergency"),
  Admissions = c(250,180,320,210,450)
)
ggplot(hospital_data,
       aes(x = Department,
           y = Admissions,
           fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Patient Admissions by Department",
       x = "Department",
       y = "Admissions") +
  theme_minimal()
ggplot(hospital_data,
       aes(x = "Admissions",
           y = Admissions,
           fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Stacked Bar Chart of Admissions",
       x = "",
       y = "Admissions") +
  theme_minimal()
ggplot(hospital_data,
       aes(area = Admissions,
           fill = Department,
           label = Department)) +
  geom_treemap() +
```

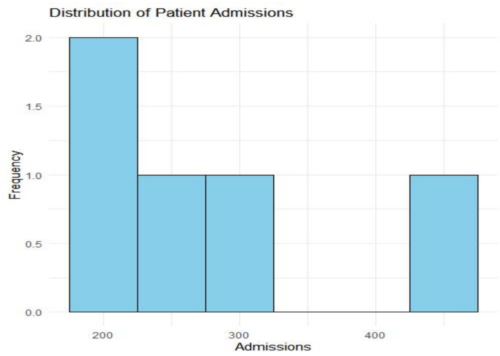
```
  geom_treemap_text(colour = "white", place = "centre") +
  labs(title = "Treemap of Patient Admissions")
ggplot(hospital_data,
       aes(x = Department,
           y = Admissions,
           fill = Department)) +
  geom_boxplot() +
  labs(title = "Admission Variability by Department",
       x = "Department",
       y = "Admissions") +
  theme_minimal()
ggplot(hospital_data,
       aes(x = Admissions)) +
  geom_histogram(binwidth = 50,
                 fill = "skyblue",
                 color = "black") +
  labs(title = "Distribution of Patient Admissions",
       x = "Admissions",
       y = "Frequency") +
  theme_minimal()
cat("\nEvaluation:\n")
cat("1. Bar Chart - Best for comparing department workload.\n")
cat("2. Stacked Bar Chart - Shows total admissions and department contribution.\n")
cat("3. Treemap - Best for showing proportion of admissions.\n")
cat("4. Box Plot - Best for admission variability (effective with larger datasets).\n")
cat("5. Histogram - Best for understanding admission distribution.\n")
cat("\nConclusion:\n")
cat("The Bar Chart is the most effective visualization for comparing department workload.\n")
cat("The Box Plot is suitable for analyzing variability.\n")
cat("The Histogram helps understand patient admission distribution.\n")
```

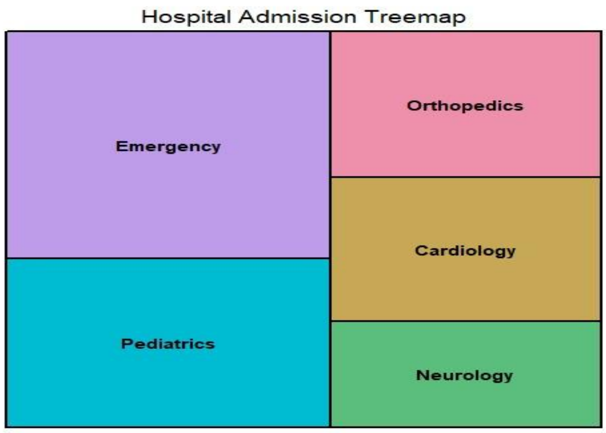
Output:











2.An online retailer has customer purchase amounts, order frequency, and productcategories. Create multiple visualizations (histogram, violin plot, box plot, density plot,bar chart). Evaluate which visualization best identifies high-value customers, spendingvariability, and purchasing trends. Recommend the most appropriate visualization forbusiness executives and explain why.

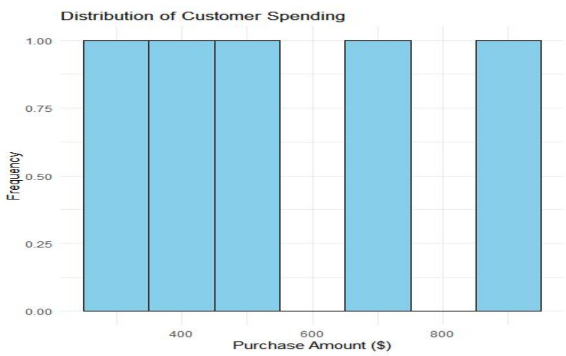
Code:

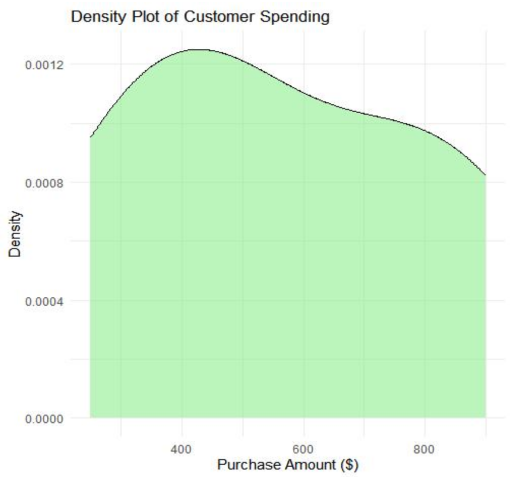
```
library(ggplot2)
customer_data <- data.frame(
  Customer = c("C1","C2","C3","C4","C5"),
  Purchase_Amount = c(250,500,750,400,900),
  Order_Frequency = c(5,8,10,6,12),
  Category = c("Electronics","Clothing","Electronics","Home","Clothing")
)
ggplot(customer_data,
       aes(x = Purchase_Amount)) +
  geom_histogram(binwidth = 100,
                 fill = "skyblue",
                 color = "black") +
  labs(
    title = "Distribution of Customer Spending",
    x = "Purchase Amount ($)",
    y = "Frequency"
```

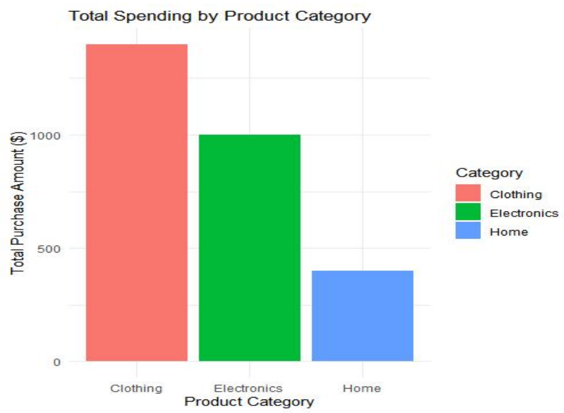
```
  ) +
  theme_minimal()
ggplot(customer_data,
       aes(x = Category,
           y = Purchase_Amount,
           fill = Category)) +
  geom_violin(trim = FALSE) +
  geom_jitter(width = 0.1,
              color = "black") +
  labs(
    title = "Customer Spending by Product Category",
    x = "Product Category",
    y = "Purchase Amount ($)"
  ) +
  theme_minimal()
ggplot(customer_data,
       aes(x = Category,
           y = Purchase_Amount,
           fill = Category)) +
  geom_boxplot() +
  labs(
    title = "Customer Spending Variability",
    x = "Product Category",
    y = "Purchase Amount ($)"
  ) +
  theme_minimal()
ggplot(customer_data,
       aes(x = Purchase_Amount)) +
  geom_density(fill = "lightgreen",
               alpha = 0.5) +
  labs(
    title = "Density Plot of Customer Spending",
    x = "Purchase Amount ($)",
    y = "Density"
  ) +
  theme_minimal()
category_sales <- aggregate(Purchase_Amount ~ Category,
                            data = customer_data,
                            sum)
ggplot(category_sales,
       aes(x = Category,
           y = Purchase_Amount,
           fill = Category)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Total Spending by Product Category",
    x = "Product Category",
    y = "Total Purchase Amount ($)"
  ) +
  theme_minimal()
```

```r
cat("\nEvaluation:\n")
cat("1. Histogram - Shows customer spending distribution.\n")
cat("2. Violin Plot - Displays spending distribution across categories.\n")
cat("3. Box Plot - Best for identifying high-value customers and spending variability.\n")
cat("4. Density Plot - Shows overall spending trend.\n")
cat("5. Bar Chart - Compares total spending by product category.\n")
cat("\nRecommendation:\n")
cat("The Box Plot is the best visualization for identifying high-value customers and spending variability.\n")
cat("The Bar Chart is the most suitable visualization for business executives because it clearly compares total spending across product categories.\n")
```

Output:







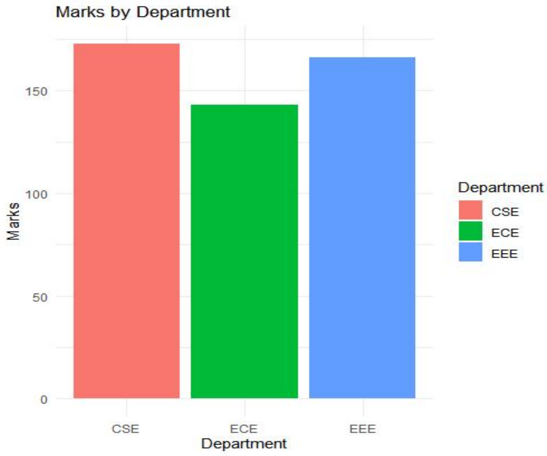
2. A university wants to compare student marks across departments and identify scoredistributions, outliers, and top-performing classes. Generate different visualizations andevaluate which charts provide the clearest insights for academic decision-making.Recommend improvements to the visualization dashboard.

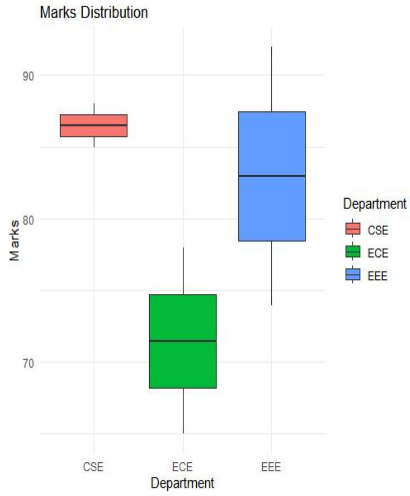
Code:

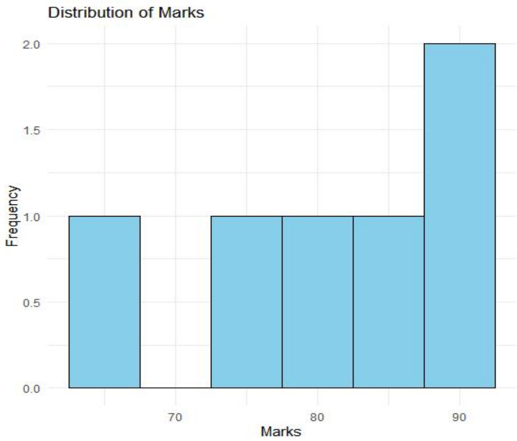
```
library(ggplot2)
library(plotly)
student_data <- data.frame(
  Student = c("S1","S2","S3","S4","S5","S6"),
  Department = c("CSE","ECE","EEE","CSE","ECE","EEE"),
  Marks = c(85,78,92,88,65,74),
  Class = c("A","A","B","B","A","B")
)
ggplot(student_data,
       aes(x = Department,
           y = Marks,
           fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title="Marks by Department",
       x="Department",
       y="Marks") +
  theme_minimal()
ggplot(student_data,
       aes(x = Department,
           y = Marks,
           fill = Department)) +
  geom_boxplot() +
  labs(title="Marks Distribution by Department",
       x="Department",
       y="Marks") +
  theme_minimal()
ggplot(student_data,
       aes(x = Marks)) +
```

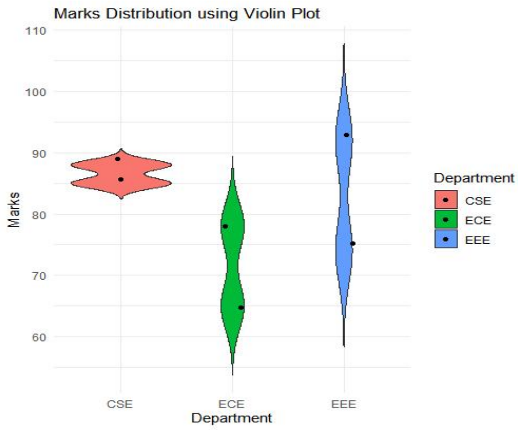
```
  geom_histogram(binwidth=5,
               fill="skyblue",
               color="black") +
  labs(title="Distribution of Student Marks",
       x="Marks",
       y="Frequency") +
  theme_minimal()
ggplot(student_data,
       aes(x = Department,
           y = Marks,
           fill = Department)) +
  geom_violin(trim=FALSE) +
  geom_jitter(width=0.1) +
  labs(title="Marks Distribution (Violin Plot)",
       x="Department",
       y="Marks") +
  theme_minimal()
ggplot(student_data,
       aes(x = Class,
           y = Marks,
           color = Department,
           size = Marks)) +
  geom_point(alpha=0.7) +
  labs(title="Top Performing Classes",
       x="Class",
       y="Marks") +
  theme_minimal()
cat("\nEvaluation\n")
cat("Bar Chart : Compare department marks.\n")
cat("Box Plot : Detect score distribution and outliers.\n")
cat("Histogram : Understand overall marks distribution.\n")
cat("Violin Plot : Shows score density.\n")
cat("Scatter Plot : Identifies top-performing classes.\n")
cat("\nRecommendation\n")
cat("Use Bar Chart + Box Plot + Scatter Plot in the dashboard.\n")
```

Output:









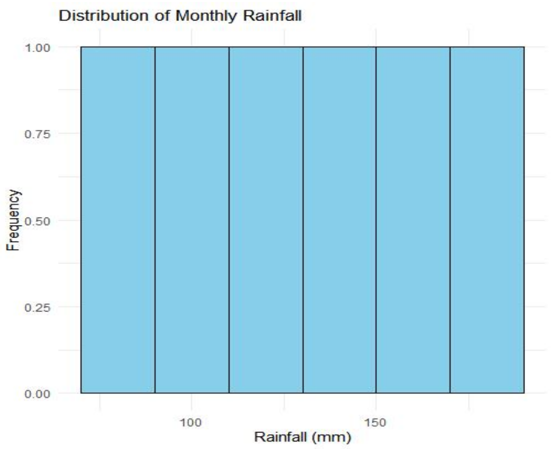
3. Meteorological data contains monthly rainfall, temperature, and humidity for multiplecities. Produce histograms, density plots, box plots, and bar charts. Evaluate whichvisualization best represents seasonal rainfall variation and climate distribution.Compare at least three visualizations and justify the most suitable one for environmentalresearchers.

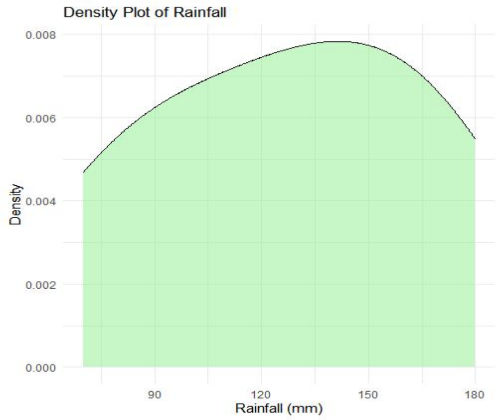
Code:

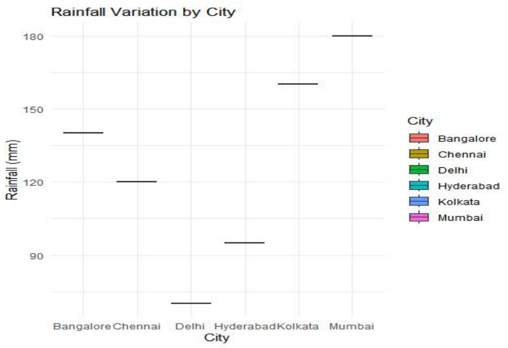
```
library(ggplot2)
climate_data <- data.frame(
  City = c("Chennai","Hyderabad","Bangalore","Mumbai","Delhi","Kolkata"),
  Rainfall = c(120,95,140,180,70,160),
  Temperature = c(34,32,28,30,36,33),
  Humidity = c(78,65,82,88,55,80)
)
ggplot(climate_data,
       aes(x = Rainfall)) +
  geom_histogram(binwidth = 20,
                 fill = "skyblue",
                 color = "black") +
```

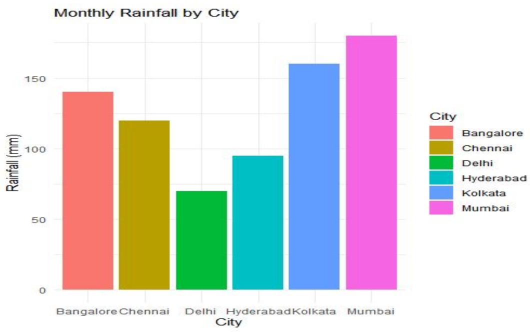
```r
  labs(title = "Distribution of Monthly Rainfall",
       x = "Rainfall (mm)",
       y = "Frequency") +
  theme_minimal()
ggplot(climate_data,
       aes(x = Rainfall)) +
  geom_density(fill = "lightgreen",
               alpha = 0.5) +
  labs(title = "Density Plot of Rainfall",
       x = "Rainfall (mm)",
       y = "Density") +
  theme_minimal()
ggplot(climate_data,
       aes(x = City,
           y = Rainfall,
           fill = City)) +
  geom_boxplot() +
  labs(title = "Rainfall Variation by City",
       x = "City",
       y = "Rainfall (mm)") +
  theme_minimal()
ggplot(climate_data,
       aes(x = City,
           y = Rainfall,
           fill = City)) +
  geom_bar(stat = "identity") +
  labs(title = "Monthly Rainfall by City",
       x = "City",
       y = "Rainfall (mm)") +
  theme_minimal()
cat("\nEvaluation\n")
cat("Histogram : Shows rainfall distribution.\n")
cat("Density Plot : Displays rainfall trend smoothly.\n")
cat("Box Plot : Best for identifying rainfall variation and outliers.\n")
cat("Bar Chart : Best for comparing rainfall among cities.\n")
cat("\nRecommendation\n")
cat("For environmental researchers, the Box Plot is most suitable because it clearly shows seasonal rainfall variation and detects outliers.\n")
cat("The Bar Chart is useful for comparing rainfall between cities, while the Density Plot provides an overall view of climate distribution.\n")
```

Output:









4. A financial institution has customer loan amounts, repayment history, income levels, andcredit scores. Develop visualizations showing loan distributions and customersegments.Evaluate which visualization best supports loan approval decisions by highlighting riskpatterns, skewness, and outliers. Recommend the visualization that should be includedin a management dashboard.

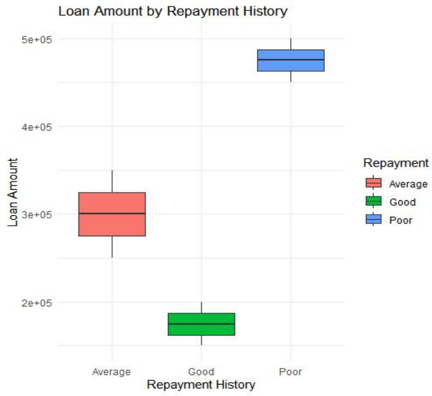
Code:

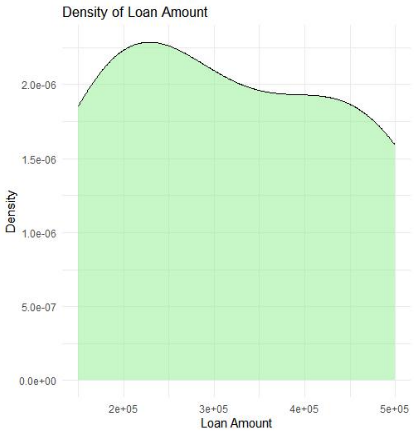
```
library(ggplot2)
loan_data <- data.frame(
  Customer = c("C1","C2","C3","C4","C5","C6"),
  Loan_Amount = c(200000,350000,150000,500000,250000,450000),
  Income = c(50000,80000,40000,100000,60000,90000),
  Credit_Score = c(720,680,750,620,700,650),
  Repayment = c("Good","Average","Good","Poor","Average","Poor")
)
ggplot(loan_data,
       aes(x = Loan_Amount)) +
  geom_histogram(binwidth = 50000,
                 fill = "skyblue",
                 color = "black") +
  labs(title = "Loan Amount Distribution",
       x = "Loan Amount",
       y = "Frequency") +
  theme_minimal()
ggplot(loan_data,
```

```
      aes(x = Repayment,
          y = Loan_Amount,
          fill = Repayment)) +
  geom_boxplot() +
  labs(title = "Loan Amount by Repayment History",
      x = "Repayment History",
      y = "Loan Amount") +
  theme_minimal()
ggplot(loan_data,
      aes(x = Loan_Amount)) +
  geom_density(fill = "lightgreen",
            alpha = 0.5) +
  labs(title = "Density of Loan Amount",
      x = "Loan Amount",
      y = "Density") +
  theme_minimal()
ggplot(loan_data,
      aes(x = Credit_Score,
          y = Loan_Amount,
          color = Repayment,
          size = Income)) +
  geom_point(alpha = 0.7) +
  labs(title = "Credit Score vs Loan Amount",
      x = "Credit Score",
      y = "Loan Amount") +
  theme_minimal()
ggplot(loan_data,
      aes(x = Repayment,
          fill = Repayment)) +
  geom_bar() +
  labs(title = "Customers by Repayment History",
      x = "Repayment History",
      y = "Number of Customers") +
  theme_minimal()
cat("\nEvaluation\n")
cat("Histogram : Shows loan amount distribution.\n")
cat("Box Plot : Identifies outliers and loan variability.\n")
cat("Density Plot : Displays skewness of loan amounts.\n")
cat("Scatter Plot : Shows risk pattern between credit score and loan amount.\n")
cat("Bar Chart : Compares customer segments based on repayment history.\n")
cat("\nRecommendation\n")
cat("The Scatter Plot is the best visualization for loan approval decisions because it highlights customer risk patterns.\n")
cat("The Box Plot helps identify outliers, while the Density Plot shows skewness in loan amounts.\n")
```

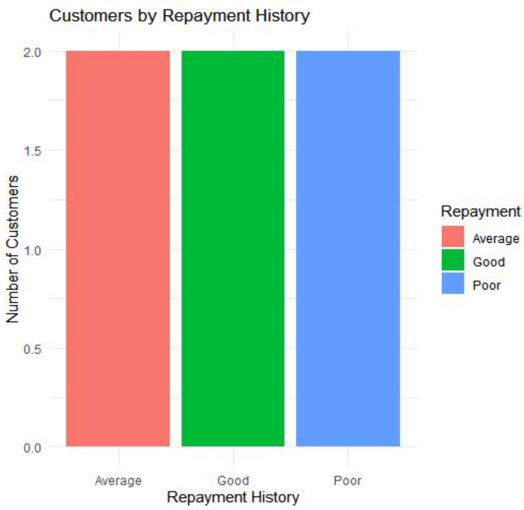
Output:


Loan Amount Distribution
1.00
0.75
0.50
0.25
0.00
Frequency
2e+05
3e+05
4e+05
5e+05
Loan Amount


Loan Amount by Repayment History
5e+05
4e+05
3e+05
2e+05
Loan Amount
Average
Good
Poor
Repayment History
Repayment
Average
Good
Poor


Density of Loan Amount
2.0e-06
1.5e-06
1.0e-06
5.0e-07
0.0e+00
Density
2e+05
3e+05
4e+05
5e+05
Loan Amount


Credit Score vs Loan Amount
5e+05
4e+05
3e+05
2e+05
Loan Amount
640
680
720
Credit Score
Repayment
Average
Good
Poor
Income
4e+04
5e+04
6e+04
7e+04
8e+04
9e+04
1e+05


Customers by Repayment History
2.0
1.5
1.0
0.5
0.0
Number of Customers
Average
Good
Poor
Repayment History
Repayment
Average
Good
Poor